Ferrofluid Negative Viscosity (Full SI Formulation)

April 14, 2026

Magnetic Definitions

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) \tag{1}$$

Symbols:

- $\mathbf{B}$: Magnetic flux density (T)
- μ_0: Permeability of free space (H/m)
- $\mathbf{H}$: Magnetic field (A/m)
- $\mathbf{M}$: Magnetization (A/m)

$$\xi = \frac{\mu_0 m H}{kT} \tag{2}$$

Symbols:

- ξ: Langevin parameter
- m: Magnetic moment (A·m^2)
- k: Boltzmann constant
- T: Temperature (K)

Relaxation Times

$$\tau_B = \frac{3\eta V}{kT} \tag{3}$$

Symbols:

- τ_B: Brownian relaxation time
- η: Fluid viscosity (Pa·s)
- V: Particle volume

$$\tau_s = \frac{\rho_s d^2}{60\eta} \tag{4}$$

Symbols:

- τ_s: Rotational relaxation time
- ρ_s: Particle density
- d: Particle diameter

Navier-Stokes Equation (Bacri Paper)

$$\rho \frac{d\mathbf{v}}{dt} = -\nabla p + \eta \nabla^2 \mathbf{v} + \frac{1}{2} \nabla \times (\mathbf{M} \times \mathbf{H}) + \mu_0 (\mathbf{M} \cdot \nabla)\mathbf{H} \tag{5}$$

Symbols:

- ρ: Fluid density (kg/m^3)
- $\mathbf{v}$: Velocity field (m/s)
- p: Pressure (Pa)
- η: Dynamic viscosity
- $\mathbf{M}$: Magnetization
- $\mathbf{H}$: Magnetic field

Note: This is Navier–Stokes with magnetic body force and torque terms :contentReference[oaicite:1]index=1

Continuity Equation

$$\nabla \cdot \mathbf{v} = 0 \tag{6}$$

Symbols:

- Incompressible flow condition

Magnetization Dynamics (Bacri)

$$\frac{d\mathbf{M}}{dt} = \mathbf{\Omega} \times \mathbf{M} - \frac{1}{\tau_B}(\mathbf{M} - \mathbf{M}_0) - \frac{1}{6\eta\phi}\mathbf{M} \times (\mathbf{M} \times \mathbf{H}) \tag{7}$$

Symbols:

- $\mathbf{\Omega}$: Fluid vorticity
- ϕ: Particle volume fraction
- $\mathbf{M}_0$: Equilibrium magnetization

Torque Balance (Bacri)

$$6\eta\phi(\omega_p - \Omega) = \mathbf{M} \times \mathbf{H} \tag{8}$$

Symbols:

- ω_p: Particle angular velocity
- Ω: Fluid vorticity

Rotational Viscosity Definition

$$\Delta\eta = \frac{3}{2}\eta\phi\frac{\Omega - \omega_p}{\Omega} \tag{9}$$

Symbols:

- $\Delta\eta$: Changed viscosity

Langevin Magnetization

$$\mathbf{M}_0 = nmL(\xi)\frac{\mathbf{H}}{H} \tag{10}$$

$$L(\xi) = \coth\xi - \frac{1}{\xi} \tag{11}$$

Symbols:

- n: Number density
- $L(\xi)$: Langevin function

Linear Approximation

$$\mathbf{M}_0 = \chi \mathbf{H}, \quad \chi = \frac{n\mu_0 m^2}{3kT} \tag{12}$$

Alternating Magnetic Field

$$\mathbf{H} = (H_0 \cos \omega t, 0, 0) \tag{13}$$

Symbols:

- H_0: Amplitude
- ω: Angular frequency

Frequency Dependent Rotational Viscosity

$$\Delta\eta(\omega) = \frac{1}{4}\eta\phi\xi^2 \frac{1 - \omega^2\tau_B^2}{(1 + \omega^2\tau_B^2)^2} \tag{14}$$

Symbols:

- Shows transition to negative viscosity

High Frequency Limit

$$\Delta\eta \sim -\frac{\phi}{6\eta}\left(\frac{\mu_0 M_s H}{\omega}\right)^2 \tag{15}$$

Symbols:

- M_s: Saturation magnetization

Experimental Reduced Viscosity

$$\eta_r = \frac{\eta(H, f) - \eta(0, 0)}{\eta(0, 0)} \tag{16}$$

Symbols:

- η_r: Reduced viscosity
- f: Frequency